

Selection and Adjunction

Abstract: This article develops a theory of syntactic selection and adjunction (based on Zeijlstra 20920) in which labeling follows directly from Merge and feature percolation, rather than from an independent labeling algorithm. Syntactic objects carry two types of formal features: independent features, which determine categorial status, and dependent features, which encode syntactic dependencies. When two objects merge, matching independent and dependent features under sisterhood do not percolate, whereas all other features do. This derives projection by selection while also accounting for adjunction: adjuncts are analyzed as elements that select their hosts while preserving their categorial status. Extending the analysis to PPs and clausal arguments yields a restricted theory of c-selection in which verbs syntactically select DPs (including nominalized clauses), whereas PPs and non-nominalized clauses function as predicate modifiers that select the verbal expressions they combine with. Finally, the proposal unifies local selection with long-distance Agree. Whereas independent features cannot percolate beyond their maximal projections, dependent features may percolate until they enter a local sisterhood relation with a matching independent feature. Upward Agree thus reduces to postponed local selection. The resulting architecture provides a unified account of labeling, selection, adjunction, and Agree.

Key words: Merge, labeling, adjunction, selection, projection, Agree.

The author declares no competing interests

1. Introduction

A central question in contemporary syntactic theory is how the categorial properties of a complex syntactic object are determined. Since the introduction of the Minimalist Program, this question has often been formulated as a question about *labeling*: when two syntactic objects merge, what determines the label of the resulting object? This question is particularly pressing under the assumption that Merge itself is essentially set formation. If Merge combines two syntactic objects without determining which of their properties project to the resulting structure, an additional labeling mechanism appears to be required.

Examples of such labeling algorithms may vary, both among proposals, or within proposals. For Chomsky (2013, 2014), going back to Chomsky (1995, 2008), different configurations receive their label on different grounds: in head-complement configurations, the head projects due to minimal search (the closest terminal node determines the label), whereas in spec-head configurations either a shared feature becomes the label (e.g., when a DP merges with something traditionally referred to as T'), or the sister of a raised element becomes the label (v labels vP when the DP moves out of its specifier as discontinuous elements cannot provide a label). Other versions of labeling algorithms have been proposed where selecting or probing features are said to project (Chomsky 1995, 2008; Adger 2003; Boeckx 2008; Chechetto & Donati 2010, 2013), or where the output of Merge is assigned a higher feature taken from a functional sequence (Adger 2013).

Those approaches are not without problems. For one, if Merge is indeed independent of labeling, then the appeal that the minimalist approach towards Merge makes (sheer set formation instead of X-Bar theory) is overshadowed by the need to postulate an additional mechanism for labeling. Moreover, while these approaches in general can capture the labeling of head-complements configurations, or labels where a specifier merges with its sister, they have notorious problems understanding the labeling of adjuncts. So far, no standard theory of the labeling of adjuncts has been formulated. In addition, those approaches that connect labeling with selection, also require a theory of featural selection. Generally, featural dependencies can be captured in terms of Agree, where a probe requires the presence of a goal. However, since selection applies in a strictly local fashion, and Agree can apply long-distance, it is not trivial to reduce selection to Agree under such approaches.

This article develops a theory, based on Zeijlstra (2020), where labeling and selection follow directly from the interaction of Merge and feature percolation. The central proposal is that syntactic objects carry two kinds of formal features: independent features, which determine

categorial status, and dependent features, which encode dependencies on other syntactic objects. When two syntactic objects merge, pairs of matching independent/dependent features that stand in a sisterhood relation are prevented from percolating to the mother node; all other features percolate. Thus, the familiar result that a selecting element projects is not stipulated but naturally derived: it follows from a more general theory of feature percolation. Moreover, the locality of selection is then also derived, as sisterhood is a strictly local relation. At the same time, the fact that dependent features can percolate in principle unlimitedly, still allows long-distance Agree.

As we will see, this theory will also be able to deal with labeling of adjuncts. Rather than taking adjuncts to be of primitive categories (e.g. adverbials or attributive adjectives), it takes them to be categorically derived. An adverb has an underlying feature set $\{[V],[uV]\}$, where $[V]$ denotes an independent verbal feature and $[uV]$ a dependent one. Effectively this states that adverbs are verbal elements that select for a verbal host. It is then the verbal feature of the adverb that projects the $[V]$ feature when the adjunct merges with a VP. Adjuncts are not selected by their host but select them.

The theory also provides a way of reinstating a restricted notion of c-selection. Although verbs need not syntactically select every type of semantic argument, the proposal argues that the syntactic arguments directly selected by verbs are DPs. PPs and certain CPs instead select the verbal predicate they combine with. Other CPs, by contrast, are analyzed as nominalized and therefore are DPs. This way c-selection can be fully re-installed: every verb selects a DP.

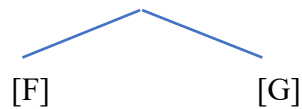
The article is organized as follows. Section 2 introduces the theory of feature percolation and labeling. Section 3 revisits three classical problems for projection and selection: the labeling of adjuncts, the (c-)selection by verbs, and the relation between selection and Agree. The next three sections discuss these in turn. Section 4 develops the analysis of adjunction and PP-modification. Section 5 turns to c-selection and s-selection, arguing that the apparent categorial flexibility of verbal arguments can be reconciled with the idea that verbs c-select DPs. Section 6 shows how Agree can be reduced to a local sisterhood relation. Section 7 summarizes the consequences of the proposal.

2. From projection by selection to feature percolation

2.1 The problem of labeling

Consider the following instance of Merge applied to an element carrying feature $[F]$ and an element carrying feature $[G]$:

(1)



If Merge is merely set formation, yielding $\{[F],[G]\}$, nothing in the operation itself determines the categorial or featural status of the resulting object. Yet, syntactic computation appears to be sensitive to the category of complex objects. A phrase headed by a verb can be selected by T, for instance, while a DP can be selected by a determiner or a verbal head. This seems to require that the top node must have some feature / category as well, even though Merge itself does not provide one.

There have been two broad ways of responding to this problem. One possibility was to dispense with labels altogether. Collins (2002, 2017), for example, argues that much of the information supplied by labels is independently recoverable from the structure. If T selects a VP, T can inspect its sister and determine whether the relevant verbal and selectional properties are present on its head.

The alternative has been to retain labels but introduce an independent labeling algorithm. Chomsky (2013, 2014), for example, proposes that different structural configurations are labeled in different ways, involving minimal search, shared features, or the properties of elements that undergo movement. Other proposals have explicitly connected projection with selection or probing.

The proposal developed here takes a third position. Labels are necessary for syntactic computation, but there is no need for a labeling operation independent of Merge. Instead, labels are the featural content that emerges from the application of Merge itself.

There are two reasons for retaining labels. First, labels make syntactic computation more efficient. A selecting head can determine the categorial status of its sister without inspecting the entire internal structure of that sister. Second, if Merge is sensitive to the formal properties of the objects to which it applies, those properties need to be available at the level of the object as a whole. A label therefore functions as a compact representation of precisely the information that is relevant for subsequent syntactic computation.

The question then arises what the proper labeling mechanism is. Must a separate labeling mechanism be postulated to apply on top of Merge, or can it follow from the features that participate in the merger what the resulting labeling would be? The latter would have the conceptual advantage that no separate labeling algorithm needs to be part of UG. However,

before spelling out how labeling can be directly derived from Merge, we first need to look into the kind of syntactic features that are involved in Merge in the first place.

2.2 Independent and dependent formal features

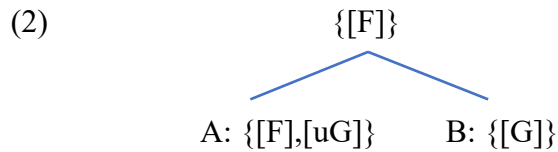
The proposal relies on a distinction between two kinds of formal features. The first are independent formal features, such as [V], [N], [D] and [T]. These determine the categorial behaviour of the objects that contain them. An element carrying [V] is verbal, an element carrying [D] is a determiner. Second, there are dependent formal features, written [uF], which encode a purely syntactic dependency on an object carrying the corresponding independent feature [F]. In that sense, they are very similar to the bullet features [•F•] that Heck and Mueller (2007, 2013) have introduced. Thus, a determiner that selects an NP carries the feature set {[D],[uN]}. The [D]-feature determines the categorial propositional behavior of the lexical item; [uN] encodes its requirement that it combine with a complement carrying [N].

This conception of formal features deliberately separates formal from semantic properties. Independent formal features do not themselves have semantic content. Rather, they determine syntactic categorial distribution. The fact that formal and semantic properties frequently correlate is a consequence of the way lexical items are learned and represented, not evidence that the two kinds of features should be identified (see Zeijlstra 2014, 2022). This means that (in)dependent features should be taken to be different from (un)interpretable features. Independent features are pure checkers, and dependent features are pure checkees. They don't feed semantics in any particular way.

The distinction also renders an independent notion of selectional features unnecessary. If an element F requires the presence of an element G, this requirement can simply be represented by [uG]. Selection is therefore not a separate, primitive operation, nor are there selectional features next to other probing features. Selection is nothing but the featureal dependency encoded on a dependent feature.

2.3 Feature percolation

Now, suppose a syntactic object A has the feature set {[F],[uG]} and merges with a syntactic object B carrying {G}.

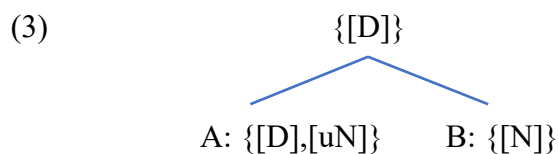


Before merger, A is a syntactic object of category F that requires the presence of a syntactic object of category G. Once A and B merge, however, that dependency has been satisfied. The resulting object should therefore still be of the categorial type of F, but it should no longer require the presence of an element of category G.

The crucial generalization is that the features involved in the featural dependency do not percolate to the mother (see also Zeijlstra 2020). There is no reason to assume the presence of a feature [G] nor of a feature [uF] on the top node. When an independent feature [F] matches a dependent feature [uF] on its sister, neither [F] nor [uF] becomes part of the label of the mother. At the same time, all other features that do not participate in such a matching relation do percolate. Note that this idea is not novel at all. It is completely reminiscent of semantic type theory (as in Heim & Kratzer 1998) or categorial grammar (Ajdukiewicz 1935, Lambek 1958, Moortgat 1997, Steedman 2000) where the same principle applies. In fact, this way minimalism can and should even be understood to be a particular kind of categorial grammar.

Informally, when two syntactic objects merge, a matching pair consisting of an independent feature [F] and a dependent feature [uF] do not percolate; all other features do percolate.

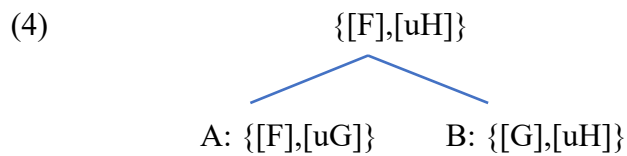
This derives the familiar result of projection by selection without stipulating that the selecting element projects. Consider the merger of a D with an NP:



The [D]-feature percolates, while [uN] and [N] do not end up on the label of the resulting syntactic object.

This differs in an important respect from traditional projection-by-selection accounts. The generalization is not that just the selecting element projects. Rather, all features percolate unless they participate in a matching dependency. This means that features belonging to the non-selecting sister can also project when they are not part of the relevant dependency. A

configuration like (4) is possible as well. Projection is therefore a consequence of feature percolation, not an independent property of heads.



2.4 One independent feature

The theory imposes a further restriction on labels: every syntactic object contains exactly one independent feature.

The motivation is distributional. If an object simultaneously belonged to two independent categories F and G, its distribution would have to be compatible with the distribution of both categories. But then it would not display the distributional restrictions that define either category. By contrast, additional specification *within* a category is possible: [D], for example, can be specified for number ([D:*num*]) without ceasing to be a determiner.

This restriction provides an important condition on Merge. If two syntactic objects both have independent features, those features cannot simply both survive on the mother. Merge therefore has to involve a matching dependency, with the corresponding independent and dependent features removed from the output label (see also Pesetsky and Torrego 2006 and Wurmbrand 2014 who reach a similar conclusions on different grounds).

The result is a highly restrictive conception of structure building. Merge is not free to combine arbitrary objects. Its application is constrained by the features on the syntactic objects involved.

The resulting picture can be summarized as follows:

- (5)
- a. Every syntactic object has exactly one independent feature.
 - b. Syntactic objects may contain any number of dependent features.
 - c. A dependent feature [uF] encodes a syntactic requirement for merger with an independent feature [F].
 - d. When [F] and [uF] occur on sister nodes, neither percolates.
 - e. All other features percolate.
 - f. A well-formed syntactic object cannot contain an unchecked dependent feature.

The familiar relation between selection and projection this way follows from a much more general theory of syntactic structure.

3. Problems of projection

The proposal is close in spirit to original projection-by-selection approaches. However, it differs from them, in that it can deal with a number of problems that other approaches have been facing. The most important ones, which I will discuss in this article (following discussion in Zeijlstra 2020), are the labeling of adjuncts, the theoretical status of c-selection, and the relation between selection and (long-distance) Agree. Here, I briefly outline why they actually have been problematic for the conjecture that selection determines projection.

3.1 Adjunction

The first problem concerns the labeling of adjunction. Consider the merger of a VP (*write a letter*) with an adverbial adjunct (*often*).

(6) [VP [AdvP often] [VP write a letter]]

Now, the resulting constituent is a VP as well, evidenced by the fact that *write a letter* and *often write a letter* are interchangeable, which diagnoses categorial identity. Yet the VP does not select the adverb. An independently existing VP is perfectly grammatical without an adjunct, and an arbitrary number of adjuncts can be added recursively. Then the question arises how the resulting structure can be a VP, if the VP itself does not select the adjunct? If projection is in some way the result of selection, it is not clear why the host should determine the label of the resulting constituent.

An additional problem concerns the status of the original VP. If the VP *often write a letter* is a VP due to the projection of the [V]-feature on *write*, it is unclear why *write a letter* would be a VP too, as it is not a maximal projection of [V]. At the same time, the diagnostics for maximal projections clearly indicate that *write a letter* is a maximal projection, as can, for instance, be shown by means of the *did (so)*-test (where the ellipsis can only target VPs).

(7) Mary often wrote a letter ...
a. ... and Bill did (so) too.

- b. ... but Bill rarely did (so)
- c. *... but Bill did (so) an email

An alternative would be to pursue that the adjunct selects the host. This is attractive because an adjunct arguably *requires* a host in a way that the host does not require the adjunct. But if selection is encoded by dependent features, the question becomes: what feature representation makes an adjunct select its host while preserving the host's categorial status? A naïve assumption would be that an adverb carries a feature set {[Adv],[uV]} but that would mean that the entire projection would end up being an AdvP rather than a VP. Then the categorial identity between *write a letter* and *often write a letter* would be lost.

3.2 C-selection and s-selection

A second problem concerns the distinction between syntactic and semantic selection. Verbs may semantically select for arguments whose syntactic categories differ:

- (8) a. to know [_{DP} the answer]
- b. to know [_{PP} about Mary]
- c. to know [_{CP} that Mary has left]

That does not mean, of course, that there cannot be any specific syntactic requirements on the arguments selected by predicates. Grimshaw (1979) already proposed a theory of selection in which semantic selection (s-selection) and syntactic selection (c(ategorial)-selection) co-exist. The reason is that the meaning of certain predicates does not always predicts its categorial restrictions. For instance, *wonder* and *ask* have similar meanings, but only *ask* allows concealed questions (Pesetsky 1991). Also, whereas *wonder* and *ask* can both select a pronominal DP complement, *inquire*, again very similar in meaning, cannot do so.

- (9) a. Mary asked {what time it was / the time}
- b. Mary wondered {what time it was / *the time}
- c. Mary {asked / wondered / *inquired} that

Naturally, one can take examples like (9) to be arguments in favour of c-selection, but nowadays the literature tries to reduce such selectional properties to other independent properties of the

lexical item (in terms of l-selection, see Pesetsky 1991 for details). However, if argument selection is not controlled by syntactic selection, how can syntax determine that it is the verb that is the selecting element? If argument selection primarily reduces to s-selection and l-selection, syntax would not be able to see the merger of a verb with an argument DP/CP/PP as the fulfilment of a syntactic requirement. Consequently, a projection by selection approach can no longer derive the fact that V, and not its argument, projects.

3.3 Selection and Agree

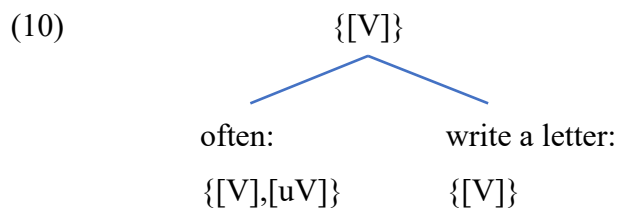
Finally, more needs to be said about selectional features. Unless one stipulates an independent set of selectional features as part of the set of formal features, selectional features should be reduced to already existing features that encode syntactic dependencies. Such features are well known and are often referred to as uninterpretable or unvalued features (independent and dependent formal features in the current proposal). These features have their ‘selectional’ requirements satisfied by means of the operation Agree. Since Chomsky (2001), however, it is generally assumed that Agree can take place in a long-distance fashion, whereas the kind of selectional requirement that is taken to underlie projection can only apply in a strictly local fashion, essentially under sisterhood or under spec-head configurations. Hence, unless it can be independently motivated why selecting or independent/dependent features must be checked locally whereas other features may also be checked on a distance, either projection should follow from something else, or selectional features should be separated from uninterpretable/dependent features, even though they both encode formal dependencies – an undesirable result from a minimalist perspective.

In the following sections, I will address these three problems and show how under the proposal pursued here they are resolved.

4. Adjunction

4.1 Adjuncts as {[X],[uX]} feature sets

The central insight of the analysis is that an adjunct modifying an X-object is itself an X-object with a dependent [X]-feature. Consider again (6). Now, suppose the host carries [V]. If the adjunct is itself represented as carrying {[V],[uV]} the derivation is straightforward:



The $[V]$ -feature of the host matches the $[uV]$ -feature of the adjunct. These two features therefore do not percolate. The independent $[V]$ -feature of the adjunct, however, has no matching dependent feature on its sister and therefore does percolate.

The resulting structure consequently contains $[V]$, but crucially the $[V]$ -feature on the mother originates from the adjunct rather than from the host. This explains two central properties of adjunction.

First, the mother is categorially identical to the host. Both the host and the combination of the adjunct and host are VPs. Second, the host itself remains a maximal projection: since it is the $[V]$ -feature of the adjunct that projects, both the host and the resulting structure are maximal projections of $[V]$. The projection line goes from the adjunct to the top node. The theory therefore derives the familiar “category-preserving” character of adjunction without requiring a special operation such as Pair-Merge or late adjunction. The analysis is closely related to categorial grammar treatments in which an adjunct has a category corresponding to X/X , and to semantic type-theoretic analyses in which an adjunct is a function from predicates of a certain type to predicates of that same type (e.g. $\langle \text{et}, \text{et} \rangle$). The important difference is that the present proposal derives the relevant categorial structure from the same feature-percolation mechanism that handles ordinary selection.

Note that treating adverbs as feature sets $\{[X],[uX]\}$ does not make them indistinguishable from ordinary X -elements. For one, it still carries a dependent feature $[uX]$. Moreover, the two independent features can still differ in their subfeatures/values. A finite verbal element, for example, can carry a richer $[V]$ -feature (e.g. $[V:\textit{fin}]$) than an adverb. Consequently, syntactic operations targeting a finite V cannot target the adjunct.

However, a potential problem lies in the fact that for every adjunct it needs to be specified what kind of element it can modify. For adverbs, one has to say that these can only modify verbal elements. That seems already fairly restricting, as it is known that most adverbs can modify all kinds of phrases in the clausal spine: VP, vP, TP, etc. Here, one might argue that such phrases still have some verbal feature that renders them members of the verbal extended projection. At the same time, it is also known that several adverbs can also modify adjectives. *An often impossible colleague* contains an adjective (*impossible*) modified by the adverb *often*.

Hence, more needs to be said or otherwise a paradox emerges: adjuncts need to carry the same independent feature as their host, but at the same time should be able to modify categorially different hosts. In order to resolve this paradoxical situation, let's focus on PPs, a class of adjuncts that is notoriously flexible.

4.2 PP-adjunction

Unlike adverbs, PPs they can modify a variety of predicates. Consider (11):

- (11) a. She was walking in the garden
b. The book on the table
c. She was afraid of the cat

At the same time, PPs do not freely modify arbitrary categories. They can modify verbal and nominal predicates, but not, for example, ordinary DP or CPs in the same way, as shown for DPs in (12) and for CPs by means of the fronting effects in (13):

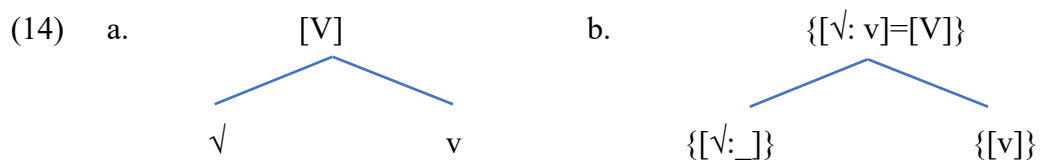
- (12) *[_{DP} [_{DP} John in the garden]] left
- (13) a. Mary thinks [*_{CP} in his wildest dreams [_{CP} that he will become president]]
b. [_{CP} that he will become president] Mary thinks
c. [*_{CP} in his wildest dreams [_{CP} that he will become president]] Mary thinks

It should be mentioned, though, that sometimes it appears that a DP can be modified by a PP. Examples are *My uncle before the tragedy is not my uncle after the tragedy* or *John in his yellow suit looked really ridiculous* (Idan Landau, p.c.). However, in the first example, the DP is actually nominalized (just as in *the Berlin of my youth*), and in the second example, *in his yellow suit* is actually a reduced relative clause. Apart from such constructions, which do not involve proper DP modification by a PP, DPs cannot be PP-modified.

It is thus not the case that PPs are completely flexible in what they adjoin to. In fact, what do adjoin to appear to be cases of verbal, nominal and (predicately used) adjective phrases, phrases that are all projected by means of a lexical head. The question then emerges what their common denominator is.

Currently, it is assumed in minimalism that lexical categories don't enter the derivation as noun, verb or adjectives but rather enter the derivation as an acategorial root that gets categorized in the syntax by merging with *v*, *n* or *a* (Marantz 1997, Embick 2021, Borer 2005a-b, Harley 2014, among others).

But rather than taking such roots to be acategorial, they can just as well be taken to be supercategorial, i.e. they carry a particular feature that in the course of the derivation gets valued for being verbal, nominal or adjectival (Harley 2014). Hence, the structure in (14)a could be replaced by the one in (14)b.



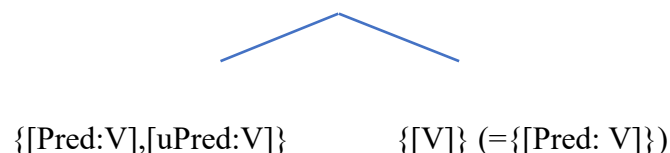
Instead of referring to a feature $[\sqrt{\cdot} \text{ _}]$, I rather refer to a superfeature [Pred(icate)] that is a superfeature of verbs, nouns and adjectives and that can be valued in the course of the derivation for being verbal, nominal or adjectival.

Now, as a next step, we can identify PPs as feature sets $\{[\text{Pred: } _], [\text{uPred: } _]\}$. Then, when a PP merges with a VP, what happens first is that the $[(\text{u})\text{Pred: } _]$ features get valued for being V, resulting in a valued $[(\text{u})\text{Pred: V}]$, which is (notationally) equivalent to a feature $[(\text{u})\text{V}]$. The [Pred: V] feature on the PP then projects a VP again.

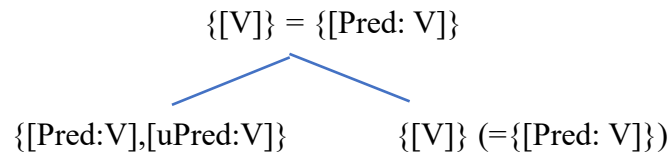
(15) Step I: Merger of the PP and VP



Step II: Valuation of the $[(\text{u})\text{Pred}]$ features:



Step III: Projection of [Pred:V]



The same way, an NP is projected when the PP merges with an NP, and an AP when the PP merges with an AP.

This analysis captures an important property of PP-adjunction: PPs are flexible precisely because their categorial identity is not initially fixed as V or N. They are predicates that select predicates.

4.3 The internal structure of PPs

The analysis has a further consequence. If a PP modifying a VP has the feature representation $\{[V],[uV]\}$, then its internal derivation must yield this representation. Suppose a preposition combines with a DP. If the DP carries $\{[D]\}$, and the resulting PP $\{[Pred],[uPred]\}$, the preposition must carry $\{[Pred],[uPred],[uD]\}$. The $[uD]$ -feature requires a DP complement. The $[Pred]$ - $[uPred]$ pair makes the resulting object capable of modifying a predicate. Thus, on this view, prepositions are structurally complex predicate-selecting elements. They are, in an some sense, adverbial elements that additionally select a DP. This analysis makes several otherwise surprising properties of prepositions natural.

For one, even though standardly a preposition can combine with a DP and subsequently with a verbal predicate, because its dependent features are unordered, it can also merge with verbal material first. This yields a direct prediction about particle-verb constructions. Consider:

(16) Mary looked up the glasses.

If *up* is represented as carrying a feature set $\{[Pred],[uPred],[uD]\}$, it can first combine with *look*, satisfying $[uPred]$ and valuing both $[(u)Pred]$ features:

$$(17) \quad \{[V],[uV]\} = \{[Pred:V],[uPred:V]\}$$



The resulting verbal object can then merge with the DP *the glasses*, satisfying [uD]’s featural requirements.

This derivation is particularly attractive because particle-verb constructions have properties intermediate between ordinary phrases and complex heads. In English and Dutch, for example, both the verbal base and the larger particle-verb constituent may participate in head movement:

- (18) a. Mary [drank up]_i [her beer]_j t_i t_j
 b. Mary drank_i [her beer]_j t_i up t_j

- (19) a. ... dat Jan zijn moeder t_i wil [op-bellen]_i ¹
 ... that Jan his mother wants up-call
 ‘... that Jan wants to call up his mother’
 b. ... dat Jan zijn moeder op- t_i wil bellen_i
 ... that Jan his mother up wants call
 ‘... that Jan wants to call up his mother’

At the same time, as is shown in the example in (20), (regular) inflectional morphology only targets the verbal base and not the entire complex verb. The same holds for cases of V-to-C movement in languages like Dutch, where only the verbal base can move to C and not the entire complex (21):

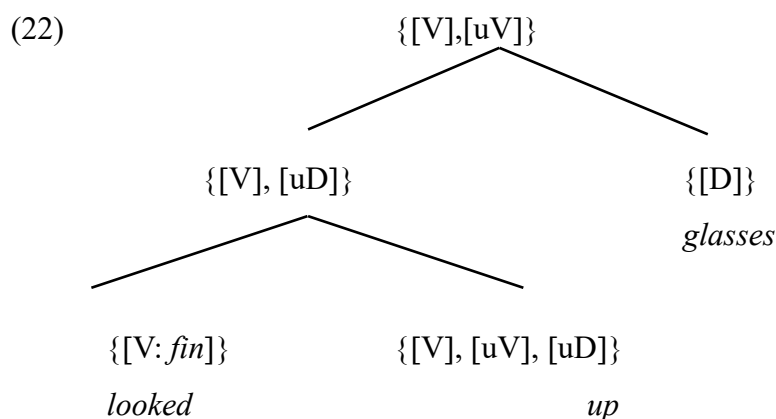
- (20) Mary looked up her glasses

- (21) a. Jan belt_i zijn moeder op t_i
 Jan calls his mother up
 ‘John calls up his mother’

¹ Taken from Neeleman (1994)

- b. *Jan opbelt_i zijn moeder t_i
 Jan up-calls his mother
 ‘John calls up his mother’

Under the present proposal, this follows from the fact that the [V]-feature of the particle—not the finite [V]-feature of the verbal base—projects to the complex object. If C targets a finite V (as in cases of V-to-C movement), it probes for [V:*fin*], and only the verbal base can be extracted:



The analysis therefore provides empirical motivation for treating selectional features being unordered with respect to each other. A system with ordered selectional requirements would incorrectly predict that the DP complement must always be merged before the verbal predicate. The existence of particle-verb structures argues against such ordering.

4.4 PP arguments versus PP adjuncts

The same analysis must distinguish PP arguments from PP adjuncts. This is important because the two are syntactically distinct despite both ultimately yielding verbal projections. Compare:

- (23) a. She is looking at the bakery.
 b. She is working at the bakery.

In the first case, *at the bakery* is an argument; in the second it is an adjunct. The distinction is visible in syntactic processes such as passivization and PP fronting, as shown below:

- (24) a. *The bakery was worked at
b. The bakery was looked at
- (25) a. At the bakery, she was working
b. ??At the bakery, she was looking

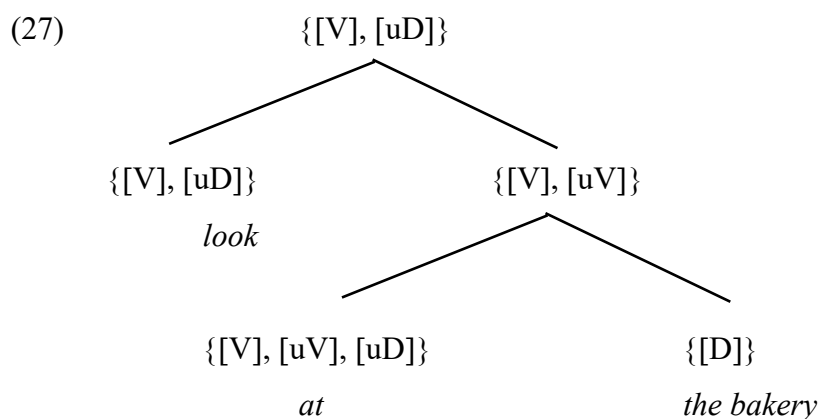
So, if both PP would be feature sets {[Pred: _], [uPred: _]}, how can the adjunct-argument distinction fall out?

The crucial assumption to be made here is that verbal predicates must saturate their arguments before they can be modified by adjuncts. This means that a PP merged with a predicate that still requires a DP argument must itself be interpreted as an argument. Once the relevant DP argument has been saturated, subsequent PP merger is adjunctal.

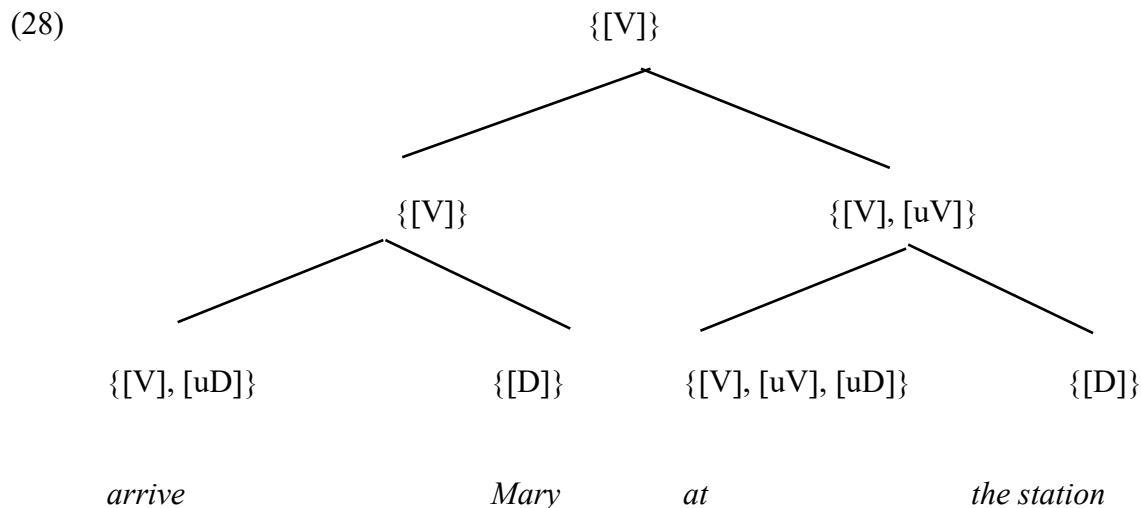
The theory encodes this distinction employing the [uD]-feature. Since the highest verbal argument is a DP (see section 5 for why clausal subjects count as DPs), a verbal predicate carries [uD] until that argument has been introduced. A PP merged with a predicate that still carries [uD] therefore satisfies an argumental configuration. A PP merged after uD has been satisfied is an adjunct. The resulting diagnostic is strikingly simple:

- (26) A PP merged with a verbal object whose label contains [uD] is an argument; a PP merged with a verbal object whose label lacks [uD] is an adjunct.

That means that in the following configuration with an unergative verb (where I ignore the presence of a DP-selecting functional head v), the PP must be an argument.



By contrast, in the next configuration, involving an unaccusative predicate, merger with a PP can only result in adjunct status of the PP, as the DP object has already satisfied the verb's argumental needs:



The argument/adjunct distinction is therefore not stipulated as a primitive difference between two kinds of PP. It follows from the position at which the PP is merged relative to the verb's unsatisfied selectional requirements.

5. Reinstalling c-selection

5.1 Why verbs appear to select heterogeneous complements

The proposal faces a natural objection. If verbs syntactically select DPs, how can one account for predicates such as *know*, which can apparently merge with DP, PP and CP arguments?

- (29)
- a. Mary knows [DP John]
 - b. Mary knows [PP about John]
 - c. Mary knows [CP that John left]

Similarly, verbs may impose lexical restrictions on otherwise identical complement categories:

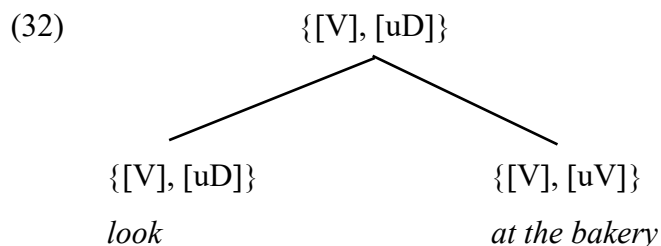
- (30)
- a. Sue relies on the bus
 - b. *Sue relies to the bus

- (31) a. Sue bristled at Sally's insult
 b. *Sue bristled to Sally's insult

Such facts, as outlined before, motivate distinctions between c-selection and semantic or lexical selection.

The present proposal does not deny semantic or lexical selection. Instead, it argues that semantic selection should not automatically be identified with the syntactic dependency that determines projection. A predicate can semantically require a particular kind of argument without that requirement being encoded as a syntactic dependent feature.

Once PPs are analyzed as predicate-selecting objects, this distinction becomes especially useful. A verb can semantically select a PP, while syntactically the PP selects the verb. Thus, the semantic argument relation and the syntactic selection relation can point in different directions. In (27), repeated in short below, the PP *at the bakery* is semantically an argument of the verb, but syntactically the verb is a host it adjoins to. That syntactically it still behaves argumentally, is due to the fact that its sister (and its mother) carry [uD].



But this means that in terms of c-selection, verbs can never syntactically select a PP argument; they can only be selected by them. But that means in turn that only DPs and CPs can be syntactically selected by verbs, rendering uniform verbal selection one step closer.

5.2 CPs as DPs or verbal modifiers

So, can verbs be said to c-select their syntactic arguments? Here, it turns out that CPs provide the crucial intermediate case. Some clausal complements behave like nominal DPs arguments; others behave more like verbal modifiers, carrying {[V],[uV]}.

The idea that clausal subjects are structurally nominal has a long history and is developed in detail by Davies & Dubinsky (1998) and Hartman (2012). On this view, a CP

First, the nominal layer associated with subject clauses is overt in some languages. In Farsi, for example, a clausal subject must be preceded by the determiner *in* ‘this’, whereas the same determiner is impossible with a CP functioning as a complement.

- Similar patterns have been reported for languages such as Russian and Greek (cf. Hartman 2012). This asymmetry suggests that the nominal structure is specifically associated with clauses occupying subject position rather than being an inherent property of all CPs.

(34) a. That Leslie arrived drunk itself put Kelly in a foul mood.
b. Kelly was angry that Leslie arrived drunk (*itself).

Agreement provides a third argument. Coordinated clausal subjects can trigger plural agreement, as in *That the president will be re-elected and that he will be impeached are equally likely*. Crucially, plural agreement cannot simply be analysed as default agreement. It therefore indicates that clausal subjects can bear ϕ -features, a characteristic property of DPs. Although other categories such as PPs, VPs and APs sometimes appear to trigger singular agreement, Hartman argues that these cases are largely restricted to copular constructions and are better analysed as involving an implicit *pro*. With ordinary lexical verbs, such non-nominal phrases

make very poor subjects. Consequently, apparent agreement with non-nominal constituents does not undermine the argument that genuine clausal subjects are nominal.

Finally, clausal subjects participate in case computation. Transitive constructions containing a clausal subject allow the object to bear accusative case just as constructions with a nominative DP subject do. If accusative is a marked case licensed in the presence of a higher nominative argument, this suggests that the clausal subject itself must count as a nominative as well.

(35) That she left surprised me/*I

More direct evidence comes from languages such as Korean, where clausal subjects overtly bear nominative case:

(36) Muess-ul mek- ul- kka- **ka** manje- i- ta.
what-Acc eat-FUT-COMP-NOM problem be- DECL
'What to eat is the problem' (Han 2005)

Taken together, determiner realization, emphatic-reflexive modification, ϕ -agreement, and case marking provide converging evidence that clauses appearing in canonical subject position are DPs rather than bare CPs.

This then may give rise to the impression that only clausal subjects are DPs and that other clausal arguments are CPs. However, this claim has been argued to be too strong, both by Kastner (2015) and by Bondarenko (2022). Both, on different grounds and in different ways, argue that certain object clauses are actually DPs while others are not. For Kastner, the latter are still CPs; for Bondarenko, they are verbal modifiers. This latter, more radical, approach would render a feature [C] obsolete. Below I briefly discuss the Bondarenko's approach, in more detail.

Bondarenko argues that non-subject clauses can be either CPs or DPs, depending on their syntactic and semantic role. In her system, thematic roles can only be satisfied by DPs. CPs, by contrast, function as modifiers, specifying the propositional content of an attitude predicate. Nominalization thus has both syntactic and semantic consequences.

To illustrate this, she provides examples like (37) from Buryat, which is a language where nominalization of clauses happens overtly.

- (37) a. Dugar [_{DP} mi:sgəi zagaha ədj-ə: g-ə:š-i:jə] han-a:
 Dugar cat.nom fish eat-pst say-part-acc think-pst
 ‘Dugar remembers (the claim/idea) that the cat ate the fish.’
- b. Dugar [_{CP} mi:sgəi zagaha ədj-ə: gəžə] han-a:
 Dugar cat.nom fish eat-pst comp think-pst
 ‘Dugar thought that the cat ate the fish.’

(37)a means that Dugar’s thought was: The cat ate the fish; not that Dugar was thinking *about* the cat eating fish. By contrast, the latter is exactly what (37)b means. This sentence lacks the reading where the propositional content of the think has been modified. What is important for our purposes that there is an asymmetry between subject clauses (which are always DPs) and non-subject clauses (at least in derived positions according to Hartman (2012), which can be structurally ambiguous between DPs and verbal modifiers. In what follows, we adopt this position.

Under the present theory, a non-nominalized CP can be analyzed as a verbal modifier {[V],[uV]}.² If the complementizer selects a TP, the complementizer can carry a further dependent T-feature: {[V],[uV],[uT]}. The CP then combines with T and yields a verbal modifier.

Nominalized CPs, by contrast, are DP. They may be represented as embedded under an overt or covert nominal shell. In that case, there is a null determiner that selects a verbal modifier and yields a DP ({[D],[V],[uV]}).

This analysis has an important consequence. If PP arguments and non-nominal CPs are verbal modifiers, then they do not need to be selected by the verb. They select the verb instead. The apparent heterogeneity of verbal complements is therefore reduced to a more restricted syntactic system: DPs are syntactically selected by verbs; nominalized CPs are DPs and can therefore also be selected by verbs; PP arguments select the verbal predicate; non-nominal CPs behave similarly to PP modifiers and select the verbal predicate as well.

The result opens up the way for reinstallment of c-selection without denying the existence of semantic or lexical selection. The only category that can be selected by verbs are DPs. The next question is then whether verbs can or must select DPs. In the next section, I will argue that they must.

² Note that this can be generalized to ({[Pred: _],[uPred: _]}) to account for the fact that nominal modifiers, like relative clauses, can also be headed by *that*-complementizers.

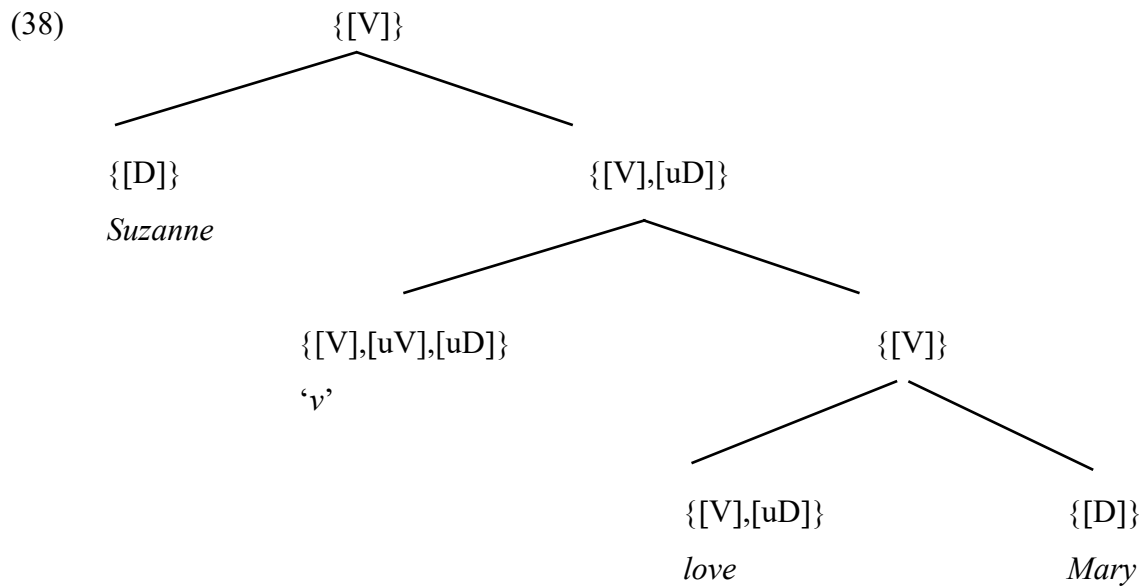
5.3 How verbs select DPs

The idea that verbs select DPs receives further support from the behavior of verbal arguments. The highest argument of a verbal predicate is consistently nominal. There are no ordinary configurations in which a PP functions as the highest argument while a DP occupies a lower argument position.

The theory captures this by assigning all verbs a [uD]-feature. An unaccusative such as *arrive* can therefore be represented as carrying a feature set {[V],[uD]}. The verb then merges with a DP and projects V.

The analysis then raises a further issue. If verbs can select DPs, why can a verb not simply carry two [uD]-features and select two DPs directly? The answer follows from the theory of features as sets. A set cannot contain two identical members. The grammar consequently cannot represent two independent DP-selection requirements on the same verbal head. How then will a second DP be selected by a verb?

Following a long tradition going back to Larsson (1989), Hale and Keyser (1993) and Kratzer (1996) on the structure of the verbal domain, the theory introduces higher verbal heads to introduce additional DP arguments. A higher v-head can be represented as {[V],[uV],[uD]}. It first combines with a VP, satisfying [uV], and subsequently combines with a DP, satisfying [uD]. The result is a verbal projection {[V]}. Note that this projection is generally referred to as vP, but there is no need to postulate an additional feature [v] next to [V]. The difference between what is canonically referred to as v and V resides fully in the fact that the former selects a VP and the latter not. Note that since T selects both 'vPs' and 'VPs', there is no reason that there would be two different kinds of selecting T-heads. The structure of a transitive predicate like *Suzanne loves Mary* is therefore as follows:

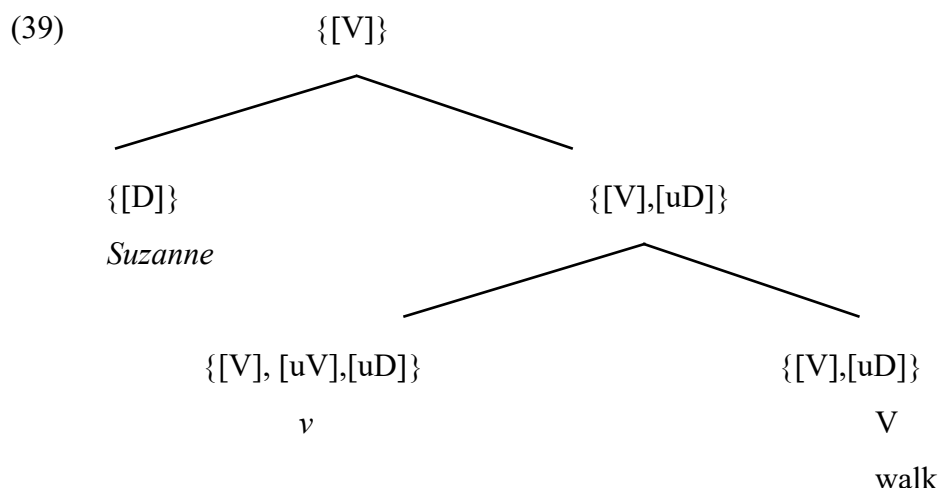


The lower V thus selects the object DP; the higher verbal head selects the VP and introduces the external argument DP.

This is not merely a restatement of familiar layered-VP structures. The present theory provides a formal reason why such structures are necessary. Since a single feature set cannot contain two identical [uD]-features, multiple DP arguments require multiple verbal heads.

This also explains why the highest verbal argument is a DP. If every verbal head ultimately requires its highest argument to be introduced by a [uD]-feature, a non-DP highest argument would leave an unchecked [uD]-feature on the resulting syntactic object.

So far, we have distinguished (di-)transitive verbs from (intransitive) unaccusatives verbs by assuming that both verbs select a DP, i.e. the verbs themselves are feature sets {[V],[uD]}, and that the transitive ones are selected by a higher verbal head that selects the second DP argument (and, similarly, ditransitive even by two verbal heads selecting a VP, each also selecting one extra argument). This still leaves open the question as to how to deal with unergative verbs. Adopting the standard assumption that unergative verbs have their argument selected by *v*, the following option suggests itself:



In (39) the verb, as in the other examples, carries a feature set {[V],[uD]}. As both V's and v's feature sets contain an identical unchecked feature [uD], the top node only contains one [uD] too. Evidence for this extra, redundant [uD] feature on V comes from the well-known empirical fact that unergative verbs may select cognate objects, as in (40) (see, among others, Massam 1990, Levin and Rappaport Hovav 1995, Mittwoch 1998, Felser and Wanner 2001).

- (40) a. I dreamed a dream
b. Mary sang a song

By contrast, unaccusatives cannot select such cognate objects.³ This then forms evidence that every verb carries a feature [uD], and consequently, verbal c-selection is fully reinstalled.

6. Local selection vs Long-Distance Agree

So far, we have seen that selection is a local operation, applying under sisterhood, involving independent and dependent features. Those independent and dependent features are very reminiscent of interpretable and uninterpretable features along the lines of Chomsky (1995, 2000, 2001), which can establish an Agree relation.

However, since Chomsky (2000) it is generally assumed that Agree can take place in a long-distance fashion, whereas the kind of selectional features that are taken to underlie projection can only apply in a strictly local fashion, essentially under sisterhood or under spec-

³ See Kuno and Takami (2004) for discussion and examples and Nakajima (2006) for an account why those cognate objects that apparently are selected by unaccusative verbs (and thus appear to form an exception to the generalization that only unergative predicates select cognate objects) are actually not arguments.

head configurations. Hence, unless it can be independently motivated why selecting independent/dependent features must be checked locally whereas other features may also be checked on a distance, either projection should follow from something else, or selectional features should be separated from uninterpretable/dependent features, even though they both encode formal dependencies – an undesirable result from a minimalist perspective.

In this light, it should be noted that Wurmbrand (2012a-b), Zeijlstra (2012) and Bjorkman & Zeijlstra (2019) have argued that the traditional directionality of Agree should be reversed, such that the goal c-commands the probe rather than vice versa. Following Zeijlstra's (2012) definition of Upward Agree, α can Agree with β iff α carries a dependent feature, β carries a matching independent feature, β c-commands α , and β is the closest suitable goal.

Independent evidence suggests, however, that agreement relations can operate in both directions (Baker 2008, see also Preminger & Polinsky 2019). In some cases, the goal could never have c-commanded the probe, favouring Downward Agree; in others, the probe could never have c-commanded the goal, favouring Upward Agree. An example of the former comes from cross-clausal long-distance agreement in Tsez, where an embedded absolutive object can control agreement on the matrix verb when it functions as a topic.

- (41) Enir [uʒā magalu b-āc'ruʎi] b-iyxo Tsez
 mother [boy bread.ABS (III) III-ate] III-know
 'The mother knows [(that) (as for the bread) the boy ate it].'

The reverse configuration is found in phenomena such as Negative Concord, where negatively marked but semantically non-negative indefinites enter into an agreement relation with a negative operator:

- (42) Gianni non ha detto niente a nessuno.
 Gianni NEG has said neg-thing to neg-body
 'Gianni didn't say anything to anybody.'

According to Zeijlstra (2004, 2022), Negative Concord involves a syntactic dependency between a single element carrying an interpretable negative feature [iNEG] and one or more elements carrying an uninterpretable negative feature [uNEG]. Here, the interpretable feature c-commands the uninterpretable features, in accordance with Upward Agree.

The directionality of Agree has consequently been subject to substantial debate. One way to avoid positing two conflicting definitions of Agree, or parametrizing its directionality across languages or feature types, is the proposal by Bjorkman & Zeijlstra (2019), following Pesetsky & Torrego (2007) and Arregi & Nevins (2012), that feature *checking* and feature *valuation* should be distinguished. Feature checking, as in Negative Concord, applies strictly top-down: the checker must c-command the checkee. Valuation, by contrast, can proceed in either direction depending on the configuration. In ordinary subject-verb agreement, for instance, the checker may also value the checkee, whereas in long-distance agreement a lower element may provide valuation.

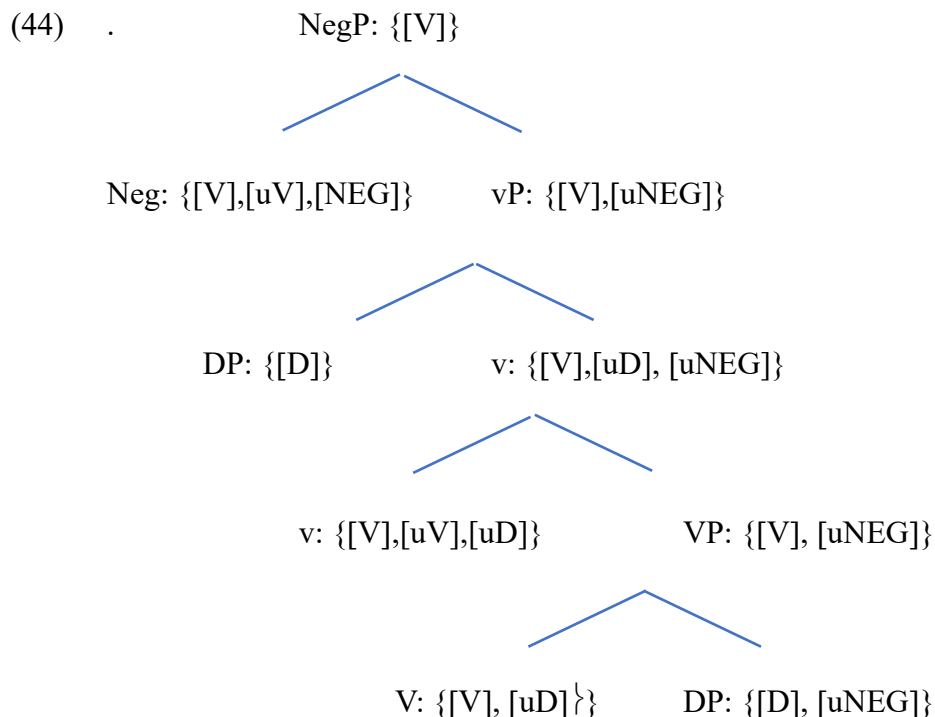
The proposal developed here derives the Upward Agree pattern independently from the behaviour of formal features. If all formal features ultimately reduce to categorial features, traditional projection lines can be understood as the percolation of independent features. Crucially, by definition such features cannot percolate beyond their maximal projection: a [V]-feature, for instance, cannot percolate beyond the VP it projects. Indeed, maximal projections can themselves be understood as the endpoints of the projection lines of independent features.

Dependent features behave differently. Nothing prevents such a feature from percolating beyond the projection in which it originates. If an element carrying [uF] does not merge directly with an element carrying the corresponding [F], [uF] must continue to percolate upwards until it encounters the appropriate independent feature. This asymmetry between independent and dependent features derives the configuration underlying Upward Agree. A dependent feature percolates upwards until it enters into a sisterhood relation with a matching independent feature, whereas an independent feature itself cannot percolate beyond its maximal projection.

On this view, Agree or more precisely, feature checking, and selection reduce to the same underlying relation. Apparent long-distance checking is simply an instance of postponed selection: a dependent feature that cannot immediately enter into a selectional relation continues to percolate until the relevant sisterhood configuration is established.

This also applies Negative Concord. A negative indefinite carrying [uNEG] need not itself be structurally adjacent to the interpretable negative operator. Instead, its [uNEG]-feature percolates upwards through the verbal structure until it reaches the projection at which it can enter into a sisterhood relation with the independent [NEG]-feature of the negative operator. Again, the apparently non-local Agree relation is reduced to a sequence of feature percolation followed by a strictly local checking relation.

- (43) a. Gianni non ha detto niente Czech
 Milan neg has said neg-thing
 ‘Gianni didn’t say anything’
 b. [TP Gianni non_[iNEG] ha detto niente_[uNEG]]



The proposal therefore derives the characteristic directionality of Upward Agree without stipulating a separate grammatical operation that searches upward through the syntactic tree. Projection of independent features is restricted by the maximal projections they define, whereas dependent features can continue to percolate until their requirements are satisfied. Consequently, whenever a dependent feature is introduced below the independent feature on which it depends, the resulting relation necessarily has the appearance of Upward Agree.

This also explains why checking dependencies such as Negative Concord systematically exhibit the relevant structural asymmetry. Their apparent non-locality does not arise because Agree itself is intrinsically non-local. Instead, the dependency is mediated by the percolation of dependent features, while the actual checking relation remains strictly local. What has traditionally been described as Upward Agree can therefore be understood as the visible consequence of a more general property of feature projection: dependent features may postpone satisfaction of their selectional requirements until they reach a configuration in which local selection under sisterhood becomes possible.

7. Conclusion

This article has developed a theory of selection and adjunction in which labeling is derived from Merge and feature percolation rather than imposed by an independent labeling algorithm.

The starting point was the observation that projection-by-selection accounts capture an important generalization but face several serious difficulties. In particular, they do not straightforwardly explain adjunction, the heterogeneous categorial status of verbal arguments, and the relation between selection and Agree.

The proposed solution is based on a distinction between independent and dependent formal features. Independent features determine syntactic category; dependent features encode requirements for matching independent features. When two syntactic objects merge, matching independent/dependent feature pairs do not percolate to the mother, while all other features do. This simple mechanism derives ordinary projection. A verb carrying a feature set $\{[V],[uD]\}$ that merges with a DP yields an object carrying $\{[V]\}$. It also derives adjunction. An adjunct modifying an X-object carries $\{[X],[uX]\}$; when it merges with X, the host's $[X]$ -feature and the adjunct's $[uX]$ -feature do not project, while the adjunct's independent $[X]$ -feature does. Consequently, the resulting object is $[X]$ and the host remains a maximal XP.

PP-adjunction motivates a further refinement. PPs are treated as predicate modifiers, with a supercategory $[Pred]$ that encompasses nominal, verbal and adjectival predicates. Prepositions themselves are represented as predicate-selecting elements with an additional $[uD]$ requirement. This analysis explains their ability to function both as arguments and adjuncts and derives their flexible behavior in particle-verb constructions. The latter, in turn, provides evidence that selectional requirements should not be ordered intrinsically.

The analysis of verbal selection then allows a restricted form of c-selection to be retained. PPs and non-nominal CPs are treated as verbal modifiers that select the predicate they merge with, whereas DPs and nominalized CPs are arguments that verbs syntactically select. Thus, V's can be said to carry a selecting feature $[uD]$. Layered verbal structures follow from the fact that a single feature set cannot encode multiple identical DP-selection requirements: additional DP arguments therefore require additional verbal heads.

Finally, the proposal derives long-distance (Upward) Agree without requiring a separate operation that searches upward through the syntactic structure. Independent features cannot percolate beyond their maximal projections, whereas dependent features continue to percolate until their requirements are satisfied. Apparent Upward Agree arises when such an independent feature originates in a higher position than the its corresponding dependent feature. This also

accounts for checking dependencies such as and Negative Concord. The resulting theory consequently reduces several apparently independent phenomena to a single architecture. Projection, selection, and adjunction are not governed by separate mechanisms. They follow from the interaction of Merge with a restricted system of formal features.

The central result can therefore be stated succinctly: Syntactic categories project not because heads project, but because matching independent and dependent features prevent themselves from percolating under sisterhood. This view preserves the computational advantages of labeling while eliminating the need for an autonomous labeling algorithm, and it provides a unified formal basis for selection and adjunction.

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